Do Post-Training Algorithms Actually Differ on Mathematical Reasoning at Scale?

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Abstract

Practitioners routinely benchmark post-training algorithms at small scale, then deploy the winner at large scale. We frame the question as a *scaling-law study of post-training algorithms*—measuring how algorithm rankings evolve as a function of model size—and show this practice is *anti-predictive* for mathematical reasoning: rankings invert as models grow. In a controlled study spanning $\sim$350 training runs across two model families (Qwen 2.5 0.5B–14B, Gemma 3 1B–12B), 8 algorithms, and 20 DPO variants with up to $N{=}5$ seeds, we find three results. **(1) Rankings invert with scale**: at $\leq$1.5B, Self-Play RFT (SP-RFT) is the strongest offline method; at 7B+, SP-RFT collapses to worst while DPO leads ($p < 0.001$). The inversion replicates in Gemma 3 and persists at Qwen 14B. **(2) Initialization dominates algorithm choice**: switching Instruct$\rightarrow$Base compresses the spread 3–15$\times$. **(3) Apparent differences have two regimes**: format compliance at small scale, architecture-specific reasoning at large scale. Together, these findings imply that small-scale algorithm selection is unreliable for mathematical reasoning, and practitioners should validate at deployment scale.

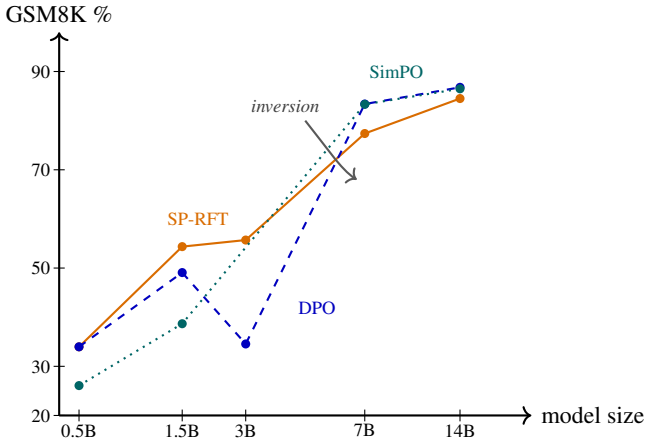

Figure 1: **A scaling-law view of post-training algorithms.** GSM8K accuracy vs. model size (log axis) for three offline post-training methods on Qwen 2.5. Algorithm *rankings* cross between 3B and 7B: SP-RFT is best at $\leq$3B but worst at $\geq$7B, while DPO and SimPO catch up and overtake. Selecting an algorithm at 1.5B is *anti-predictive* of the choice at deployment scale. The inversion replicates on Gemma 3 (1B–12B; Section 4.2).

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1 Introduction

Suppose you are choosing a post-training algorithm for mathematical reasoning. You benchmark Self-Play RFT (SP-RFT), DPO, and SimPO at 1.5B parameters: SP-RFT wins by 5 pp over DPO. You select SP-RFT, scale to 7B, and deploy—only to discover that SP-RFT is now the *worst* method by 6 pp, while DPO has silently become best. Your small-scale selection was not just uninformative; it was anti-predictive (Figure 1).

We frame this as a scaling-law study of post-training algorithms: rather than chase a single best loss function, we measure how algorithm rankings evolve as a function of model size, treating the curves themselves—not the point estimates—as the object of study.

This is not a hypothetical. It is the central empirical finding of this paper, replicated across two model families and confirmed with formal statistical testing ($p < 0.001$, $N{=}5$ seeds). The practitioner's dilemma is real: every published comparison operates at a single scale, yet deployment happens at a different one. If rankings invert, the entire methodology of algorithm benchmarking at small scale is called into question.

Post-training of large language models—adapting a pretrained model for tasks such as mathematical reasoning—has become the decisive stage separating a capable base model from a useful one (Ouyang et al., 2022; Touvron et al., 2023). Many algorithms compete for adoption: online RL methods like GRPO (Shao et al., 2024), offline preference methods like DPO (Rafailov et al., 2023), and a growing family of variants—SimPO (Meng et al., 2024), KTO (Ethayarajh et al., 2024), ORPO (Hong et al., 2024)—each claiming improvements over prior work. DeepSeek-R1 (DeepSeek-AI, 2025) demonstrated that pure RL post-training can elicit sophisticated reasoning, further expanding the design space. Yet each paper evaluates on different models, datasets, and scales, making cross-method comparison unreliable. The few existing studies (Xu et al., 2024; Spangher et al., 2025) cover limited methods at a single scale—they cannot detect the ranking inversions we observe.

We address this gap through a large-scale, controlled comparison using a unified framework where all algorithms share identical infrastructure. Our thesis is sharp: **on mathematical reasoning, post-training algorithm rankings invert with scale, making small-scale selection anti-predictive.** We emphasize that this is a task-specific phenomenon: on multiple-choice general benchmarks (ARC-C, HellaSwag, WinoGrande), algorithms differ by $<$1.2 pp at all scales (Section 4.4; Appendix T), though generative benchmarks remain untested. The evidence for mathematical reasoning unfolds in three findings:

1. **Rankings invert with scale—from Instruct checkpoints, across architectures.** On Qwen 2.5 Instruct, SP-RFT leads offline methods at $\leq$3B but collapses to worst at 7B, where DPO (83.4% $\pm$0.56) and SimPO (83.3% $\pm$1.79) dominate ($p < 0.001$ vs. SP-RFT, $N{=}5$). On Gemma 3 IT, the same pattern holds: SP-RFT $>$ DPO at 1B and 4B, but DPO $\geq$ SP-RFT at 12B (87.4% vs. 86.7%).
2. **Initialization explains most of the variance.** Replacing Instruct with Base initialization collapses the inter-algorithm spread from 15.7 pp to 1.07 pp at 1.5B and from 6.00 pp to 1.82 pp at 7B (a 3–15$\times$ compression). The dominant source of inter-algorithm variance is how each loss function interacts with prior alignment, not intrinsic properties of the objectives.
3. **Apparent differences have two regimes: format compliance at small scale, architecture-specific reasoning at large scale.** At 1.5B, the 19.3 pp GSM8K spread collapses 36$\times$ to 0.54 pp on MATH and 41$\times$ to 0.47 pp on general benchmarks—algorithms differ only in format compliance. At larger scales ($\geq$4B), MATH reveals genuine reasoning differences, but these depend on the (algorithm, architecture) pair: DPO gains +9 pp on Gemma 12B MATH while SimPO gains +13 pp on Qwen 14B MATH. Both regimes yield the same practical conclusion: small-scale benchmarking is unreliable.

These findings are based on $\sim$323 training runs on Qwen 2.5 (0.5B–14B), supplemented by 27 runs on Gemma 3 (1B–12B), spanning 8 algorithms, 20 DPO variants, 4 evaluation domains (GSM8K, MATH, ARC-C/HellaSwag/WinoGrande), and up to $N{=}5$ seeds with formal statistical testing. The implication for practitioners is direct: validate at deployment scale, not prototype scale. The implication for the research community is that designing new loss functions at small scale—the dominant paradigm—may be selecting for the wrong properties.

2 Related Work

Post-training algorithms and comparative studies. The RLHF pipeline (Ouyang et al., 2022) uses PPO (Schulman et al., 2017); GRPO (Shao et al., 2024) simplifies this via group-normalized rewards. Wu et al. (2025) show a theoretical correspondence between GRPO and DPO; our empirical results show this correspondence is scale-conditioned (SGRPO leads DPO by +8.9 pp at 1.5B but trails by -2.8 pp at 7B), illustrating that scale-dependent empirical separations can coexist with their scale-free theoretical equivalence. DPO (Rafailov et al., 2023) bypasses reward modeling, spawning IPO (Azar et al., 2024), SimPO (Meng et al., 2024), KTO (Ethayarajh et al., 2024), ORPO (Hong et al., 2024), and others. We provide the first controlled evaluation of 20 DPO variants under identical conditions. Xu et al. (2024) and Ivison et al. (2024) cover only 2–3 algorithms. Spangher et al. (2025) benchmark 17 algorithms at a single scale—they cannot detect ranking inversions. Saeidi et al. (2025) evaluate DPO variants without multi-seed runs or multiple comparison correction. Our finding that variant rankings are scale-dependent is reminiscent of Lucic et al. (2018), who found that GAN variant rankings change across experimental conditions. Critically, none of these studies evaluate across multiple model families, leaving the question of architectural generality unanswered. Our Gemma 3 validation addresses this gap. Tan et al. (2025) study scaling behaviors of RL post-training; our work complements theirs by showing that *relative* rankings—not just absolute performance—are scale-dependent.

3 Methods and Experimental Setup

Framework. Our framework provides a single codebase where all 51 algorithms share identical model loading (HuggingFace, `flash_attention_2`), data pipelines, distributed training (DeepSpeed ZeRO-3 (Rajbhandari et al., 2020), vLLM (Kwon et al., 2023) for RL rollouts), and checkpointing. Each algorithm implements a `BaseAlgorithm` interface requiring only the loss function, ensuring the *only* difference between runs is the loss computation. We organize algorithms into: **(1) SP-RFT; (2) Online RL** (GRPO variants); **(3) Offline preference** (DPO (Rafailov et al., 2023), IPO (Azar et al., 2024), SimPO (Meng et al., 2024), KTO (Ethayarajh et al., 2024), and 16 additional variants); **(4) Hybrid** (ORPO (Hong et al., 2024), SPIN (Chen et al., 2024)). The 20-variant taxonomy is in Appendix A. **SGRPO** (Self-play GRPO) is our online-RL baseline: a token-level GRPO (Shao et al., 2024) variant that performs on-policy rollouts against the same self-play prompt set used by the offline methods, scoring each rollout with exact-match correctness and applying group-normalized advantages without an explicit KL term (an ablation in Appendix P shows KL has <0.2 pp effect). It serves as a controlled online counterpart to SP-RFT on identical data and prompts. Two further GRPO variants—**GSPO** (sequence-level importance ratio, robust on MoE-style routing noise) and **CISPO** (clipped ratio used as a detached weight on the log-probability)—are reported as ablations in Appendix E; main-table "online RL" numbers refer to SGRPO unless noted.

Evaluation. GSM8K (Cobbe et al., 2021) (1,319 test problems, exact-match, greedy decoding). MATH (Hendrycks et al., 2021) (5,000 problems, 4-shot) at 1.5B and general-domain benchmarks (ARC-C, HellaSwag, WinoGrande) at both 1.5B and 7B. All evaluations use the LM Evaluation Harness (Gao et al., 2024). We report means $\pm\sigma$ and perform Welch's t-tests with Bonferroni correction where indicated.

Qwen 2.5 (primary). Qwen 2.5 (Qwen Team, 2025) Instruct at 0.5B, 1.5B, 3B, 7B, and 14B. Our experiments comprise: (1) core comparison of 8 algorithms across 4 scales ($\sim$70 runs, N=5 seeds for core methods at 7B), with a full-FT/LoRA factorial at both 3B and 7B; (2) DPO variant taxonomy at 1.5B (20 variants $\times$ 5 seeds = 100 runs) and a stratified subset of 5 at 7B (15 runs); (3) 14B deployment-scale validation with SP-RFT, DPO, SimPO (N=3 seeds each); (4) ablations ($\sim$57 runs), including Base-vs-Instruct initialization and hyperparameter sensitivity; (5) a fixed-dataset control training 1.5B on 14B self-play data (6 runs, Section 4.5); (6) LR sweeps at 1.5B, 3B, and 7B (54 runs, Appendix M–O); (7) a gold-data SFT control using MetaMathQA at 1.5B and 7B (6 runs, Section 4.5). Total Qwen-side: $\sim$323 training runs, $\sim$740 GPU-hours on H100s (Appendix I); together with the Gemma 3 runs below the grand total is $\sim$350 runs.

Gemma 3 (cross-architecture validation). To test whether our findings generalize beyond Qwen, we train SP-RFT, DPO, and SimPO on Gemma 3 (Gemma Team, 2025) IT at 1B, 4B, and 12B with 3 seeds each (27 runs). This uses the same GSM8K data and evaluation protocol; 12B uses LoRA, matching the Qwen 7B setup.

Data. All algorithms share the same data pipeline to eliminate data-source confounds. For each prompt in GSM8K's 7,473 training problems, we sample 16 responses from the *target model* at temperature 1.0 and score each with exact-match correctness. DPO/SimPO/IPO receive (chosen, rejected) pairs from the highest/lowest scoring responses. SP-RFT trains on the *chosen* trajectories from this same self-play dataset—not gold responses—so only the loss function differs between methods.[1] Because SP-RFT trains on model-generated chosen trajectories rather than gold or expert data, it is functionally *rejection sampling fine-tuning* (RFT) (Yuan et al., 2023); we use the SP-RFT label to distinguish it from standard SFT on gold data (Section 4.5). This distinction matters mechanistically: at small scale, where few model samples are correct, RFT on rare successes yields large gains; at large scale, where the model already solves most problems, RFT encounters a ceiling effect while DPO's contrastive signal can still extract improvements (see Section 5).

Design rationale. Data is generated *at each scale*, matching how practitioners train (Shao et al., 2024; DeepSeek-AI, 2025; Chen et al., 2024). Dataset quality therefore correlates with model scale; we address this confound with a fixed-dataset control (Section 4.5). **Training.** AdamW, cosine LR (10^{-6}) with 10% warmup, 3 epochs, batch size 8. LoRA (Hu et al., 2022) ($r=16$, $\alpha=32$) at $\geq$7B; full FT at smaller scales. BF16 on H100 80GB; $\beta=0.1$, SimPO $\gamma=0.5$. **Learning rate.** We sweep LR for SP-RFT and DPO at 1.5B, 3B, and 7B (Appendix M–O). The default 10^{-6} is optimal at 7B; higher rates degrade performance monotonically, with DPO more sensitive than SP-RFT (DPO collapses to 16% at 10^{-5} vs. SP-RFT at 53%). The ranking inversion (DPO $>$ SP-RFT) is preserved at independently optimized LRs for each algorithm. At 1.5B, SP-RFT leads at every LR tested, with DPO collapsing at $\geq 5 \times 10^{-6}$.

4 Results

4.1 Finding 1: Rankings Invert with Scale

Table 1 presents our main results on Qwen 2.5. At **1.5B**, SGRPO (online RL) tops all methods at $58.0\% \pm 0.57$. Among offline methods, SP-RFT leads ($54.4\% \pm 0.59$), followed by IPO, KTO, DPO, and SimPO. At **3B**, the default LR (10^{-6}) produces catastrophic format non-compliance (base model: 62.9% flexible but 10.7% strict-match). A dedicated LR sweep (Appendix O) identifies the optimal LR for each algorithm.[2] Critically, **SP-RFT leads at every LR tested**, preserving the small-scale ranking. The 3B results therefore reinforce rather than undermine the scaling story: the SP-RFT $>$ DPO ordering holds at $\leq$3B under both shared and independently optimized LRs.

The ordering *inverts* at **7B** (Figure 2). With $N=5$ seeds for SP-RFT, DPO, and SimPO, DPO reaches $83.38\% \pm 0.56$ and SimPO $83.32\% \pm 1.79$; a Welch's t-test fails to detect a significant difference ($p = 0.95$; note that a high p-value indicates failure to reject the null, not confirmation of equivalence—our sample size may lack power to resolve small gaps). The critical finding: SP-RFT drops from best offline method to *worst* at 7B ($77.38\% \pm 1.11$), separated by $\sim$6 pp from DPO/SimPO ($p < 0.001$ after Bonferroni correction over the family of 10 pairwise tests at 7B; $\alpha = 0.005$, see Appendix K). This is the SP-RFT-to-preference inversion: the method that wins at small scale loses at large scale.

Disentangling scale from LoRA. The headline curve mixes full FT at $\leq$3B with LoRA at $\geq$7B—a practitioner-realistic choice (full FT is infeasible at 7B+ on common hardware) that confounds scale with training method. A 2$\times$2 factorial at 7B trains SP-RFT/DPO/SimPO under *both* regimes: DPO leads under Full FT (84.5%) just as under LoRA (83.4%), and the inversion appears in both cells (Appendix D), so **scale**, not LoRA, drives the inversion; the mixed-method curve remains a residual limitation that full-FT runs at every scale could close.

[1] **Effective dataset overlap.** A prompt enters the SP-RFT set if at least one of its 16 samples is correct, and enters the DPO/SimPO/IPO set if it has both a correct and an incorrect sample. The two sets coincide on the dominant central regime where pass@16 is neither 0 nor 1; they diverge only in the tails (very low or very high base accuracy). At 0.5B/1.5B/7B Instruct on Qwen 2.5 the asymmetric drops are small ($< 5\%$ of prompts) and affect both methods directionally similarly, leaving the inversion intact; per-scale effective-dataset sizes will be released with the per-seed outputs.

[2] The 3B Qwen 2.5 Instruct checkpoint exhibits a pathological sensitivity to learning rate: the default 10^{-6} is too conservative for full FT at this scale. Table 1 reports the best-LR results; see Appendix O for the full sweep. SP-RFT recovers to 55.7% at 10^{-5} while DPO peaks at 34.6% (5×10^{-6}).

Table 1: GSM8K exact-match accuracy (%) across scales (Qwen 2.5). 1.5B: 3-seed means $\pm\sigma$; 7B: 5-seed means for SP-RFT/DPO/SimPO, 3-seed for IPO/KTO. Best per scale in **bold**. [b]Single seed. [c]Epoch-2 checkpoint. [d]Best of 3 epochs. [a]62.9% flexible but 10.69% strict-match. The 3B column reports best-LR results from the dedicated 3B sweep (Appendix O); at the shared default LR 10^{-6}, the 3B numbers are SP-RFT 18.35, DPO 14.44, SimPO 6.90, included so the column is comparable to the rest of the table; the SP-RFT>DPO ordering at 3B holds at *every* LR tested.

Algorithm	0.5B	1.5B	3B (full FT)	7B (LoRA)
Base Model	19.56	31.16	10.69[a]	75.82
SP-RFT	$33.97_{\pm 0.0}$ (+14.4)	$54.36_{\pm 0.59}$ (+23.2)	$\mathbf{55.70}_{\pm 1.09}$ (+45.0)	$77.38_{\pm 1.11}$ (+1.6)
DPO	$33.97_{\pm 0.0}$ (+14.4)	$49.08_{\pm 0.61}$ (+17.9)	$34.55_{\pm 4.52}$ (+23.9)	$\mathbf{83.38}_{\pm 0.56}$ (+7.6)
IPO	$\mathbf{34.50}_{\pm 0.3}$ (+14.9)	$52.24_{\pm 0.22}$ (+21.1)	—	$80.39_{\pm 0.92}$ (+4.6)
KTO	$33.81_{\pm 0.0}$ (+14.3)	$51.15_{\pm 1.77}$ (+20.0)	—	$80.24_{\pm 0.16}$ (+4.4)
SimPO	$26.08_{\pm 0.0}$ (+6.5)	$38.67_{\pm 1.78}$ (+7.5)	—	$83.32_{\pm 1.79}$ (+7.5)
SGRPO	32.45 (+12.9)	$\mathbf{58.00}_{\pm 0.57}$ (+26.8)	8.11[c] (−2.6)	80.59[d] (+4.8)

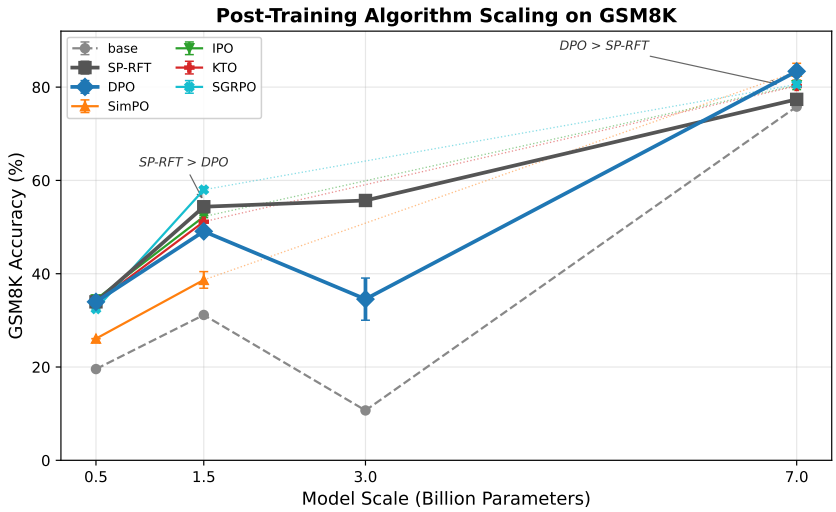

Figure 2: GSM8K accuracy across model scales (Qwen 2.5). At 7B, no significant difference is detected between DPO and SimPO ($p = 0.95$, N=5), while SP-RFT separates by ∼6 pp ($p < 0.001$). The SP-RFT-to-preference **ranking inversion** between ≤3B and 7B is the paper's central finding.

4.2 Cross-Architecture Validation and Deployment Scale

Gemma 3 (1B–12B). A natural objection is that the ranking inversion may be a Qwen-specific artifact. To test generality, we train SP-RFT, DPO, and SimPO on Gemma 3 IT (Gemma Team, 2025) at 1B, 4B, and 12B (3 seeds each, 27 runs). The pattern mirrors Qwen. At **1B**, SP-RFT dominates (26.7%), DPO falls to 21.5%, and SimPO collapses to 4.1%. At **4B**, the same ordering holds with wider gaps: SP-RFT (75.3%) ≫ DPO (57.8%) ≫ SimPO (40.0%). At **12B**—the critical test—DPO (87.4%) overtakes SP-RFT (86.7%), while SimPO (75.1%) remains behind. This is the same **SP-RFT-to-DPO inversion** observed in Qwen: SP-RFT wins at small scale, DPO wins at large scale.

Two differences from Qwen are worth noting. First, SimPO does not converge to DPO at the largest scale in Gemma, suggesting that the DPO/SimPO parity observed at Qwen 7B may be architecture-specific. Second, the inversion point appears later in Gemma (12B vs. 7B), which may reflect differences in the pretrained model's formatting capabilities. Despite these nuances, the central finding—that SP-RFT-to-preference rankings invert with scale—replicates across architectures.

Qwen 14B. To extend the scaling curve beyond 12B, we train SP-RFT, DPO, and SimPO on Qwen 2.5-14B-Instruct with 3 seeds each (9 runs). The inversion persists: DPO (86.8% ±0.23) >

Table 2: Cross-architecture validation and deployment scale on GSM8K (%). Gemma 3: 3 seeds each; 12B uses LoRA. Qwen 14B: 3 seeds each, LoRA. The SP-RFT-to-DPO inversion replicates across architectures and persists at 14B.

	Gemma 1B		**Gemma 12B**		**Qwen 14B**	
Algorithm	**Mean**	$\pm\sigma$	**Mean**	$\pm\sigma$	**Mean**	$\pm\sigma$
SP-RFT	**26.7**	—	86.7	0.29	84.5	0.19
DPO	21.5	—	**87.4**	0.26	**86.8**	0.23
SimPO	4.1	—	75.1	4.88	86.5	0.68
SP-RFT−DPO gap	*+5.2 pp*		*−0.7 pp*		*−2.2 pp*	

Table 3: Base vs. Instruct initialization at 1.5B on GSM8K (%). The spread collapses from 15.7 pp to 1.07 pp; SimPO reverses from worst to best. This compression generalizes to 7B (6.00 pp → 1.82 pp; Appendix Q).

Algorithm	**From Base**	**From Instruct**	Δ
SP-RFT	60.80	$54.36_{\pm 0.59}$	+6.4
DPO	61.33	$49.08_{\pm 0.61}$	+12.2
SimPO	**61.87**	$38.67_{\pm 1.78}$	+23.2
Spread	*1.07 pp*	*15.7 pp*	

SimPO (86.5% ±0.68) > SP-RFT (84.5% ±0.19) > Base (79.5%). Within the Qwen family, the DPO–SP-RFT gap *narrows* from 6.0 pp at 7B to **2.2 pp** at 14B, indicating that SP-RFT partially closes the gap at larger scale. On general benchmarks (ARC-C, HellaSwag, WinoGrande), the spread collapses to <1.2 pp, consistent with the format compliance finding.

Notably, MATH evaluation at 14B reveals a different pattern: SimPO (39.1%) dramatically outperforms DPO (25.8%) and SP-RFT (23.0%), contrasting with Gemma 12B where DPO dominates MATH (50.2% vs. SimPO 32.7%). This architecture-dependent reversal on harder tasks further underscores that algorithm rankings cannot be extrapolated across models or tasks.

Table 2 presents the combined cross-architecture and deployment-scale results.

4.3 Finding 2: Initialization Dominates Algorithm Choice

Why do rankings invert? We hypothesize that algorithms interact differently with the prior alignment baked into Instruct checkpoints, and that removing this alignment should eliminate the differences.

Table 3 tests this at 1.5B. Starting from Base, all methods improve (SP-RFT: +6.4 pp, DPO: +12.2 pp, SimPO: +23.2 pp) and the spread collapses from 15.7 pp to just **1.07 pp**. SimPO—worst from Instruct—becomes best from Base, suggesting its reference-free nature makes it most sensitive to initialization.

Critically, this finding generalizes to 7B (Appendix Q): from Base, the spread collapses from 6.00 pp to 1.82 pp. The ranking from Base does *not* simply reproduce the 1.5B pattern: at 1.5B Base, SimPO leads (61.87%) followed by DPO (61.33%) and SP-RFT (60.80%); at 7B Base, SP-RFT leads (82.49%) followed by SimPO (80.82%) and DPO (80.67%). Thus, a residual ranking shift persists even from Base—but the key observation is that these differences are compressed to ≤1.82 pp, compared to 6–16 pp from Instruct. Initialization is the dominant source of inter-algorithm variance, even if it does not fully explain it.

DPO's sensitivity to Base initialization is theoretically expected (KL regularization anchors to a reference that cannot format math), but the convergence extends to reference-free methods (SimPO) and pure SP-RFT, so the prior alignment state—not just reference-model mechanics—is the primary source of inter-algorithm variance. Practically, most real-world post-training starts from Instruct, where algorithm choice does produce measurable differences; the Base ablations at both scales reveal that those differences are largely an initialization-loss interaction.

Table 4: Cross-task comparison at 1.5B: GSM8K vs. MATH. The 19.3 pp spread collapses to 0.54 pp (36× compression).

Algorithm	MATH	GSM8K
SP-RFT	**27.04**	54.36
SimPO	26.72	38.67
KTO	26.64	51.15
SGRPO	26.58	**58.00**
DPO	26.50	49.08
Spread	*0.54*	*19.33*

Table 5: MATH accuracy (%) across scales. At ≥4B, DPO leads in Gemma but SimPO leads in Qwen 14B.

	Gemma 3			Qwen
	1B	**4B**	**12B**	**14B**
Base	16.6	34.4	41.3	16.8
SP-RFT	13.7	34.3	41.0	23.0
DPO	16.4	**42.4**	**50.2**	25.8
SimPO	1.1	39.2	32.7	**39.1**

4.4 Finding 3: Two Regimes—Format Compliance, Then Architecture-Specific Reasoning

Tables 4 and 5 compare GSM8K and MATH results. At 1.5B, the 19.3 pp GSM8K spread collapses to **0.54 pp** on MATH—a 36× compression. Rankings shift: SGRPO drops from 1st to 4th, SimPO rises from last to 2nd. On general-domain benchmarks (ARC-C, HellaSwag, WinoGrande), the spread collapses further to **0.47 pp** (41×; Appendix G). At small scale, algorithms differ only in format compliance, not reasoning.

But this picture changes at scale. Table 5 shows MATH accuracy across Gemma 3 scales. At 1B, the raw spread is 15.5 pp, but this is driven entirely by SimPO's catastrophic collapse to 1.1%; excluding SimPO, Base/SP-RFT/DPO cluster within 2.9 pp. At 4B, DPO pulls ahead by +8.1 pp over SP-RFT. At 12B, DPO reaches 50.2%—nearly **+9 pp over base and +17 pp over SimPO**. The spread among non-pathological methods *increases* with scale, the opposite of the compression observed at 1.5B.

Critically, this pattern is *architecture-dependent*. At Qwen 14B, **SimPO** dominates MATH (39.1% ±2.2) over DPO (25.8% ±0.4) by +13 pp—the reverse of Gemma. Meanwhile, general benchmarks (ARC-C, HellaSwag, WinoGrande; Appendix T) show <2 pp spread at all scales and both architectures. **Important caveat:** these are multiple-choice or perplexity-based benchmarks that evaluate log-likelihood ranking, not open-ended generation. Since post-training algorithms primarily alter the generative distribution, the absence of differences on MC benchmarks does not rule out inversions on generative tasks (e.g., AlpacaEval, MT-Bench). Our claim is narrower: the inversion is documented on mathematical reasoning; whether it extends to other *generative* tasks remains open. The two-regime picture is: at small scale, algorithms differ shallowly (format); at large scale, algorithms differ deeply (reasoning), but *which algorithm wins depends on architecture and task*. Both regimes make small-scale extrapolation unreliable.

4.5 Controls: Disentangling Data Quality and RFT Artifacts

Fixed-dataset control. Because self-play data is generated by each model, dataset quality correlates with model size—the most important confound in our design. To disentangle data quality from model scale, we train 1.5B Qwen 2.5 Instruct on self-play data generated by the *14B* model—holding data quality fixed at the highest available level while keeping the learner at the scale where SP-RFT dominates. If the SP-RFT–DPO ranking at 1.5B were purely a data-quality artifact, providing 14B-quality data should flip the ranking in DPO's favor.

SP-RFT still beats DPO by +5.1 pp ($58.38\% \pm 0.49$ vs. $53.32\% \pm 0.70$, 3 seeds each), confirming that model scale—not data quality—is the primary driver of the ranking inversion. Both algorithms benefit comparably from higher-quality data (SP-RFT: +4.0 pp, DPO: +4.2 pp), leaving the gap stable (5.06 pp vs. 5.28 pp with own data).

Gold-data SFT control. Because SP-RFT trains on self-play trajectories (functionally RFT), a natural objection is that the ranking inversion merely reflects RFT's ceiling effect at scale, not a property of SFT *per se*. To test this, we train standard SFT on MetaMathQA (Yu et al., 2024)—a high-quality, externally generated dataset of 240K GSM8K-related problems with expert solutions (3 seeds each at both 1.5B and 7B).

Table 6: Fixed-dataset control: 1.5B on *14B* self-play data (GSM8K %). SP-RFT still leads by +5.1 pp.

Algorithm	**Mean**	σ
SP-RFT	**58.38**	0.49
DPO	53.32	0.70
Gap	*+5.06 pp*	

Table 7: Gold-SFT control: standard SFT on MetaMathQA (GSM8K %). SFT advantage collapses from +12 pp to ≈0 pp with scale.

	1.5B		**7B**	
Method	**Mean**	σ	**Mean**	σ
Gold-SFT	**61.05**	0.45	83.32	0.28
SP-RFT	54.36	0.59	77.38	1.11
DPO	49.08	0.61	**83.38**	0.56

Tables 6–7 report both controls side by side. At **1.5B**, Gold-SFT dominates all methods at $61.05\% \pm 0.45$, outperforming DPO by +12.0 pp. At **7B**, Gold-SFT reaches $83.32\% \pm 0.28$—**converging to DPO** ($83.38\% \pm 0.56$, gap: -0.06 pp). The SFT advantage collapses from +12 pp at 1.5B to statistical parity at 7B, regardless of whether the training data is self-generated or externally sourced. This **falsifies the RFT-artifact hypothesis**: even with high-quality gold data that eliminates the self-play ceiling, SFT's advantage over DPO still vanishes at scale.

4.6 Algorithm Selection Under Scale Shift

Given that small-scale rankings are anti-predictive, what should a practitioner do when target-scale validation is too expensive? Over the cross-scale ranks in Tables 1–2 we report two summary criteria: *worst-case rank* $R_{\mathrm{wc}}{=}\max_N r_a^{(N)}$ and *deployment rank* $R_{\mathrm{dep}}{=}\mathrm{mean}_{N\geq 7\mathrm{B}}\, r_a^{(N)}$. Values: SP-RFT ($R_{\mathrm{wc}}{=}4, R_{\mathrm{dep}}{=}3.5$), DPO $(4, 1)$, SimPO $(5, 2)$. DPO Pareto-dominates SP-RFT on R_{dep} at equal R_{wc}, and dominates SimPO on both; Gemma 3 replicates the order. **DPO is therefore the rank-stable bet**—best from 7B onward across both architectures, lowest seed variance ($\sigma{=}0.56$ at 7B vs. SimPO's 1.79). Caveat: at 14B DPO ($86.78{\pm}0.23$) and SimPO ($86.48{\pm}0.68$) overlap within seed noise, so R_{dep} should be read with bootstrap rank intervals (Appendix G) rather than as a hard ordering. Empirically the DPO–SP-RFT gap turns positive past an architecture-specific crossover (Qwen between 3B and 7B; Gemma between 4B and 12B), so use DPO when deployment scale is within $2\times$ of that crossover; we avoid a parametric $\Delta(N)$ form because the within-Qwen gap *narrows* from $+6.0$ pp at 7B to $+2.2$ pp at 14B. The mechanistic basis (Section 5) is that SP-RFT hits a ceiling once base accuracy is high, while contrastive objectives still extract gradient from incorrect trajectories.

4.7 Practitioner Recommendations

Our findings yield actionable guidance:

1. **Validate at deployment scale, architecture, and task.** Rankings at $\leq$3B do not predict 7B+ behavior. Moreover, MATH rankings at Gemma 12B (DPO wins) reverse at Qwen 14B (SimPO wins). This is the paper's most important practical message.
2. **DPO is a strong default on GSM8K.** DPO ranks near the top at both 1.5B and 7B in Qwen, overtakes SP-RFT at 12B in Gemma, and leads at 14B. It has the lowest seed variance ($\sigma = 0.56$ at 7B) and requires no special infrastructure. However, its advantage does not transfer uniformly to harder tasks (MATH).
3. **Distrust variant sweeps at small scale.** The complete 1.5B→7B ranking inversion among DPO variants means that small-scale variant selection is unreliable. Prefer vanilla DPO unless you can afford to validate at target scale.
4. **Budget for multiple seeds at scale.** SimPO matches DPO in mean accuracy at 7B but with $3.2\times$ higher variance ($\sigma = 1.79$). Single-seed comparisons are misleading.

5 Discussion

Why rankings invert: a mechanistic sketch. The Base-vs-Instruct ablation (Section 4.3) identifies one mechanism: ranking inversions arise partly from differential interaction between loss functions and prior alignment. We conjecture that at small scale, the Instruct model's limited capacity means

that SP-RFT's direct supervision provides the clearest gradient signal for format acquisition, while preference methods waste capacity modeling relative preferences between correct and incorrect solutions when the model can barely produce either. A complementary lens comes from viewing SP-RFT as RFT (Section 3): at 1.5B, where the model rarely produces correct solutions, RFT on the few successful trajectories yields large gains precisely because the baseline is low. At 7B, the Instruct model already solves most GSM8K problems correctly, so RFT encounters a ceiling effect: training on filtered correct solutions offers diminishing returns, while DPO's contrastive signal—which leverages *both* correct and incorrect trajectories—can still extract fine-grained reasoning improvements that teacher-forcing cannot provide. Critically, however, this RFT ceiling is *not the whole story*: the gold-data SFT control (Section 4.5) shows that even standard SFT on external expert data (MetaMathQA) converges to DPO at 7B (83.32% vs. 83.38%), despite dominating at 1.5B by +12 pp. The inversion is therefore not merely an artifact of self-play data saturation, but a genuine property of how supervised and contrastive objectives interact with model scale.

What drives accuracy in mathematical reasoning. In decreasing order of leverage: model scale ($\sim$50 pp), initialization ($\sim$15 pp), training paradigm (SP-RFT vs. preference: $\sim$6 pp at 7B), and loss function choice within the preference family ($<$1 pp for DPO vs. SimPO on GSM8K; up to 5.8 pp across DPO variants at 7B). On harder reasoning tasks (MATH), a new dimension emerges: algorithm-architecture interaction. DPO gains +9 pp on Gemma 12B MATH while SimPO gains +13 pp on Qwen 14B MATH—a complete reversal. This suggests that loss functions interact with architecture-specific representations in ways that are invisible on format-sensitive tasks like GSM8K.

Methodological caveats at 7B+. Four design choices interact: the fixed LR 10^{-6} is below the standard LoRA recommendation but an ablation rules out under-training (Appendix M); SGRPO was trained without explicit KL but adding λ_{KL}=0.01 yields $+0.15$ pp (Appendix P); the 7B full-FT factorial (Appendix D) confirms the inversion under both regimes; and the effective batch size of 8 prompts is conservative for DPO, which already wins at 7B regardless. Online RL ablations (SGRPO, GSPO, CISPO) are in Appendix E.

6 Conclusion

On mathematical reasoning, post-training algorithm rankings invert with scale: the method that wins at 1.5B loses at 7B, a finding replicated across Qwen 2.5 (up to 14B) and Gemma 3 (up to 12B) and confirmed with formal statistical testing ($p < 0.001$, N=5). On multiple-choice general benchmarks, algorithms differ by $<$1.2 pp at all scales—the inversion is task-specific within our evaluation suite. The inversion persists at 14B (DPO–SP-RFT gap of 2.2 pp), though the within-family gap narrows from 6.0 pp at 7B, suggesting SP-RFT partially closes the gap at larger scale. A fixed-dataset control confirms this is scale-driven: training 1.5B on 14B-quality data preserves the SP-RFT–DPO gap (5.1 pp vs. 5.3 pp with own data). Initialization—not the loss function—explains most algorithmic differences (3–15$\times$ spread compression when starting from Base). At small scale, residual differences largely reflect format compliance (36$\times$ spread compression from GSM8K to MATH at 1.5B), but at large scale MATH reveals genuine reasoning differences that are both architecture- and algorithm-dependent.

For mathematical reasoning, the dominant paradigm of small-scale benchmarking + extrapolation is unreliable. Practitioners should validate at deployment scale; the research community should invest in how algorithms interact with scale, initialization, and task structure, rather than designing new loss functions at a single small scale. Whether this extends to other post-training objectives (instruction following, safety) is open—our MC-only general-benchmark control limits direct extrapolation to generative tasks.

Limitations. Two model families (Qwen 2.5 14B, Gemma 3 12B) on mathematical reasoning; on multiple-choice general benchmarks the spread is $<$1.2 pp at all scales (Appendix T) but those are not generative, so transfer to AlpacaEval/MT-Bench, instruction following, safety, or creative writing remains open. Self-play data quality covaries with scale; the controls in Section 4.5 confirm the inversion is not a data-source artifact, but extending them to every scale would strengthen the claim. The 20-variant DPO sweep uses shared β=0.1, LR= 10^{-6}; per-variant tuning could surface differences our plug-and-play protocol misses. Code, configs and per-seed JSONs release upon acceptance.

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A Full DPO Variant Results

Table 8: Taxonomy of 20 DPO variants evaluated in this study, organized by primary modification to the DPO objective. All evaluated at 1.5B on GSM8K with 5 seeds.

Variant	Primary Modification	Ref.	Year
DPO	Baseline logistic loss	Y	2023
IPO	Squared loss regularization	Y	2024
SimPO	Length-normalized, no ref	N	2024
KTO	Unpaired binary feedback	Y	2024
Hinge	Hinge loss replacement	Y	2024
CPO	Contrastive preference	N	2024
ORPO	SFT-integrated preference	N	2024
RDPO	Robust divergence penalty	Y	2024
CDPO	Conservative DPO	Y	2024
BetaDPO	Adaptive β scheduling	Y	2024
CalDPO	Calibrated reward margin	Y	2024
DPOP	Pref.-weighted optimization	Y	2024
ODPO	Online DPO	Y	2024
EXO	Exponentiated gradient	Y	2024
AlphaPO	Alpha-divergence pref.	N	2024
APO	Anchored preference	Y	2024
SPPO	Self-play preference	Y	2024
RobustDPO	Dist. robust DPO	Y	2024
GPO	Generalized pref. opt.	Y	2024
FocalPO	Focal loss preference	Y	2024

Table 9: Full DPO variant results at 1.5B scale on GSM8K (5 seeds). Δ is the accuracy difference from vanilla DPO. † indicates statistical significance after Bonferroni correction ($p < 0.0026$). Cohen's d measures practical effect size.

Variant	Category	Mean (%)	Std	Δ DPO	p	d
ORPO	Reference-free	53.89	0.60	+4.12	0.0134	2.48
DPOP	Weighting/calibration	53.62	1.80	+3.85	0.0189	1.88
SPPO	Data augmentation	52.60	1.45	+2.84	0.0521	1.49
IPO	Alternative divergence	52.36	0.75	+2.59	0.0614	1.53
CalDPO	Weighting/calibration	52.16	1.58	+2.40	0.0934	1.22
CPO	Reference-free	51.43	1.57	+1.67	0.2187	0.85
GPO	Alternative divergence	50.98	2.41	+1.21	0.4370	0.52
RobustDPO	Robustness	50.89	2.13	+1.12	0.4442	0.51
CDPO	Robustness	50.72	1.89	+0.96	0.4916	0.46
KTO	Unpaired	50.48	2.11	+0.71	0.6211	0.33
Hinge	Alternative divergence	49.87	2.43	+0.11	0.9449	0.05
DPO	Vanilla	49.76	2.27	–	–	–
ODPO	Data augmentation	48.37	3.20	−1.39	0.4525	−0.50
APO	Data augmentation	48.07	2.73	−1.70	0.3175	−0.68
EXO	Data augmentation	47.52	2.92	−2.24	0.2145	−0.86
FocalPO	Weighting/calibration	47.04	3.27	−2.73	0.1686	−0.97
RDPO	Alternative divergence	46.79	2.47	−2.97	0.0835	−1.25
BetaDPO	Weighting/calibration	45.75	4.31	−4.02	0.1145	−1.17
AlphaPO	Reference-free	41.68	5.13	−8.08	0.0204	−2.04
SimPO†	Reference-free	38.23	2.12	−11.54	0.0000	−5.25

Statistical notes. Significance threshold: Bonferroni-corrected $\alpha = 0.05/19 = 0.0026$. Only SimPO ($p < 10^{-4}$) passes, and it is significantly *worse*. ORPO ($p = 0.013$) and DPOP ($p = 0.019$) would be significant uncorrected but fail after correction. Our 5-seed design provides 80% power to detect effects ≥ 1.0 pp (Cohen's $d \geq 0.8$).

At 7B: point-estimate rankings reverse. We evaluate a stratified subset of 5 variants at 7B (3 seeds each), selected to span the 1.5B ranking (1st, 2nd, 7th, 13th, 15th). ORPO—1st at 1.5B—collapses to worst at 7B (−4.79 pp vs. DPO). EXO—15th at 1.5B—rises to the top (+1.00 pp). No variant reaches Bonferroni-corrected significance at either scale. The practical conclusion: since variant rankings are statistically unresolvable at both scales and their point estimates reverse, small-scale variant sweeps provide no reliable signal for large-scale behavior.

Shared-hyperparameter design. All 20 variants use the same LR (10^{-6}) and $\beta = 0.1$. This is a deliberate choice: the 20-variant sweep evaluates *default-hyperparameter transferability*—whether variants provide gains "out of the box" with shared settings—not absolute algorithmic capability ceilings. This matches how practitioners typically deploy variants: using the original DPO paper's defaults. Per-variant tuning (e.g., optimizing β for IPO's squared loss or KTO's unpaired feedback) could reveal differences that our shared-hyperparameter design masks. Our conclusion (variants are unresolvable) applies to the plug-and-play setting; per-variant tuning could surface real differences that shared hyperparameters mask.

Table 10: Stratified DPO variant evaluation at 7B (3 seeds each). Variants were selected to span the 1.5B ranking (1st, 2nd, 7th, 13th, 15th) to test rank-order predictiveness. Vanilla DPO: 83.38% (N=5). Point-estimate rankings reverse: the 1.5B leader (ORPO) is the 7B laggard; the 1.5B underperformer (EXO) rises to top. No variant reaches Bonferroni-corrected significance at either scale.

Variant	Mean (%)	Std	Δ DPO	p	1.5B Rank
EXO	**84.38**	0.96	+1.00	0.277	15th
ODPO	83.88	1.31	+0.49	0.654	13th
GPO	83.40	1.05	+0.02	0.984	7th
Vanilla DPO	83.38	0.56	—	—	12th
DPOP	80.92	1.18	−2.46	0.085	2nd
ORPO	78.59	1.04	−4.79	0.013	1st

B Training Configurations

Table 11: Shared training configuration across all experiments.

Parameter	Value
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.95$, $\epsilon = 10^{-8}$)
Weight decay	0.01
LR schedule	Cosine with 10% linear warmup
Effective batch size	8 prompts (1×8 grad. accum.)
Training epochs	3
Precision	BF16 mixed precision
Distributed strategy	DeepSpeed ZeRO Stage 3
Hardware	$8\times$ NVIDIA H100 80GB HBM3
Sequence length	512 (math), 1024 (code)

Table 12: Algorithm-specific hyperparameters.

Algorithm	Learning Rate	Key Hyperparameters	GPUs/Run
SP-RFT	1×10^{-6}	–	1
DPO	1×10^{-6}	$\beta = 0.1$	1
SimPO	1×10^{-6}	$\beta = 0.1$, $\gamma = 0.5$	1
IPO	1×10^{-6}	$\beta = 0.1$	1
KTO	1×10^{-6}	$\beta = 0.1$	1
SGRPO	1×10^{-6}	$\lambda_{\text{KL}} = 0^*$, $n = 4$	2
GSPO	1×10^{-6}	$\lambda_{\text{KL}} = 0$, $n = 4$	2
CISPO	1×10^{-6}	$\lambda_{\text{KL}} = 0$, $n = 4$, clip $= [0.8, 1.2]$	2

*Following the default GRPO recipe. See Appendix P for an ablation with $\lambda_{\text{KL}} = 0.01$.

For models at 7B/12B and for the 3B factorial, we apply LoRA (Hu et al., 2022) with rank $r = 16$, $\alpha = 32$, applied to all linear layers. The core comparison at $\leq$3B/4B uses full fine-tuning.

C Framework Engineering

Scaling online RL to 7B required resolving three critical infrastructure issues: **(1) Ray GPU over-subscription.** Fractional `num_gpus` values in Ray caused GPU over-subscription deadlocks during multi-worker rollout generation; we enforce integer GPU allocation per worker. **(2) vLLM memory leak.** Rollout output objects were not explicitly freed between batches, causing a slow memory leak that triggered OOM after $\sim$200 steps at 7B; explicit deletion with `gc.collect()` resolved this. **(3) Epoch-transition OOM.** Optimizer state accumulated across epoch boundaries; offloading optimizer state to CPU between epochs eliminated the failure.

D LoRA Factorial

E Online RL Explorations

SGRPO (online RL) leads at 1.5B ($58.0\% \pm 0.57$, +3.6 pp over SP-RFT) at $\sim6\times$ the compute cost. At 7B, SGRPO achieves 80.59% (best of 3 epochs), placing it above SP-RFT but below DPO/SimPO. Adding KL regularization ($\lambda_{\text{KL}} = 0.01$) yields only +0.15 pp (Appendix P), confirming that the online–offline gap is genuine rather than a regularization artifact. **Caveat:** SGRPO uses the same effective batch size (8 prompts) and fixed 3-epoch schedule as offline methods; online RL typically benefits from larger batches and dynamic rollout budgets, so our SGRPO results may underestimate online RL's true ceiling. SGRPO was not evaluated at 12B or 14B due to its $\sim6\times$ compute cost. Token-level SGRPO outperforms sentence-level GSPO by 12–18 pp; GSPO and CISPO results are below.[3]

[3] GSPO achieves 39.58% at 1.5B; CISPO fails catastrophically (5.23% at 1.5B, 0.15% at 0.5B).

Table 13: 2×2 factorial disentangling scale from LoRA (Qwen 2.5). The inversion appears under both Full FT and LoRA at 7B: DPO (84.5%) > SimPO (83.6%) > SP-RFT (79.8%) under Full FT, mirroring the LoRA ranking. 7B LoRA values: 5-seed means $\pm\sigma$; Full FT: single seed.

		3B	7B
Full FT	Base	10.69	75.82
	SP-RFT	**18.35**[b] (+7.7)	79.83 (+4.0)
	DPO	$14.44_{\pm 2.01}$ (+3.8)	**84.46** (+8.6)
	SimPO	6.90[b] (−3.8)	83.62 (+7.8)
LoRA	Base	10.69	75.82
	SP-RFT	18.04 (+7.4)	$77.38_{\pm 1.11}$ (+1.6)
	DPO	16.00 (+5.3)	$\mathbf{83.38}_{\pm 0.56}$ (+7.6)
	SimPO	6.90 (−3.8)	$83.32_{\pm 1.79}$ (+7.5)

SGRPO at 3B. SGRPO at 3B achieves 4.93% after epoch 1 and 8.11% after epoch 2, both below the 10.69% baseline. On-policy rollouts compound the format compliance crisis: poorly formatted generations receive no reward, preventing the policy from learning formatting.

SGRPO at 7B. SGRPO achieves 80.59% (best of 3 epochs; 80.59/80.29/79.91), above SP-RFT but below DPO/SimPO. Slight accuracy degradation at epochs 2–3 suggests overfitting to the reward signal.

GSPO and CISPO. GSPO (sentence-level policy gradient) achieves 39.58% at 1.5B and 19.86% at 0.5B—12–18 pp below token-level SGRPO, demonstrating that credit assignment granularity is a high-impact choice within online RL. CISPO fails catastrophically at all scales (0.15% at 0.5B, 5.23% at 1.5B).

F SimPO Variance Analysis

SimPO exhibits the highest seed variance of any method at any scale ($\sigma = 1.79$ at 7B vs. DPO's $\sigma = 0.56$). Across 5 seeds, SimPO's per-seed accuracy ranges from 81.50% to 85.82% (range: 4.32 pp), while DPO ranges from 82.79% to 84.23% (range: 1.44 pp)—a 3× difference. SimPO's two highest and two lowest seeds cluster in pairs, consistent with a bifurcation in the optimization landscape. This bimodal pattern is absent in DPO. SimPO computes length-normalized log-probabilities without a reference model, lacking the implicit regularization of DPO's reference-anchored gradient. The practical consequence: SimPO's "best seed" appears to outperform DPO (85.82 vs. 84.23), but no significant difference is detected under proper multi-seed evaluation ($p = 0.95$).

G Additional Figures and Tables

Table 14: General-domain evaluation at 1.5B: ARC-Challenge (acc_norm), HellaSwag (acc_norm), WinoGrande (acc). The 19.3 pp GSM8K spread collapses to 0.47 pp (41× compression).

Algorithm	**ARC-C (%)**	**HellaSwag (%)**	**WinoGrande (%)**	**Avg (%)**
Base Model	46.76	68.23	62.90	59.30
SP-RFT	46.16	68.16	62.90	59.08
DPO	46.67	68.38	62.67	59.24
IPO	46.76	68.36	62.67	59.26
KTO	**47.01**	68.25	63.06	59.44
SimPO	46.08	**68.37**	62.59	59.01
SGRPO	46.67	68.23	**63.54**	**59.48**
Spread	*0.93*	*0.22*	*0.95*	*0.47*

Table 15: General-domain evaluation at 7B (LoRA). Despite a 6.0 pp GSM8K spread, the general-domain spread is just 0.71 pp.

Algorithm	**ARC-C (%)**	**HellaSwag (%)**	**WinoGrande (%)**	**Avg (%)**
Base Model	55.20	80.48	71.11	68.93
SP-RFT	55.29 (+0.09)	80.45 (−0.03)	70.72 (−0.39)	68.82 (−0.11)
DPO	54.86 (−0.34)	**80.58** (+0.10)	70.72 (−0.39)	68.72 (−0.21)
IPO	**55.55** (+0.35)	80.37 (−0.11)	**71.35** (+0.24)	**69.09** (+0.16)
KTO	55.03 (−0.17)	80.39 (−0.09)	70.72 (−0.39)	68.71 (−0.22)
SimPO	54.44 (−0.77)	80.53 (+0.05)	70.17 (−0.95)	68.38 (−0.56)
Spread (trained)	*1.11 pp*	*0.21 pp*	*1.18 pp*	*0.71 pp*

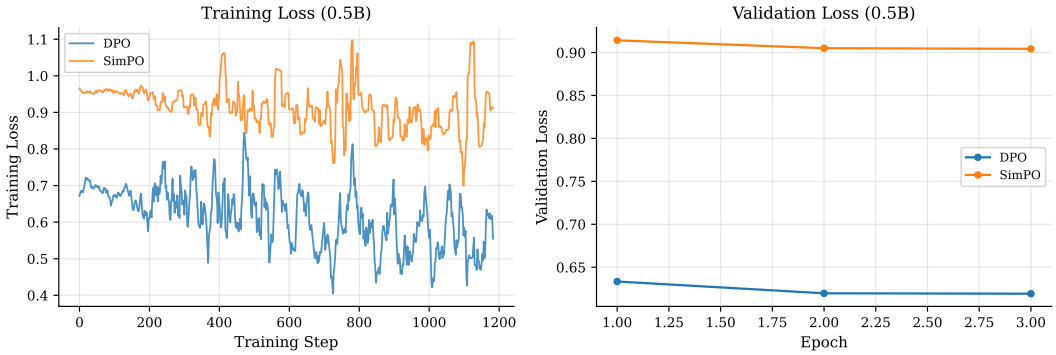

Figure 3: Training dynamics comparison at 0.5B scale. **Left:** training loss over steps (smoothed). **Right:** validation loss over epochs.

H Format Compliance Analysis

Table 16 reports strict-match and flexible-extract accuracy. The *format compliance gap* (strict − flexible) quantifies whether a model produces well-formatted answers.

Table 16: Strict-match vs. flexible-extract accuracy on GSM8K. At 3B, the format compliance gap ranges from −26.2 to −33.7 pp across trained models, confirming format non-compliance drives the low strict-match scores.

Scale	**Algorithm**	**Strict (%)**	**Flexible (%)**	**Gap (pp)**
1.5B	SP-RFT	54.66	55.57	−0.9
	DPO	51.18	49.81	+1.4
	IPO	52.39	52.31	+0.1
	KTO	51.71	51.18	+0.5
	SimPO	43.44	43.06	+0.4
3B (full FT)	Base	10.69	62.93	−52.2
	SP-RFT	18.04	44.28	−26.2
	DPO	16.45	42.91	−26.5
	IPO	16.83	45.72	−28.9
	KTO	15.31	44.50	−29.2
	SimPO	6.90	40.56	−33.7
7B (LoRA)	SP-RFT	76.42	80.29	−3.9
	DPO	83.85	84.99	−1.1
	IPO	81.35	83.09	−1.7
	KTO	80.21	82.49	−2.3
	SimPO	85.82	80.97	+4.9

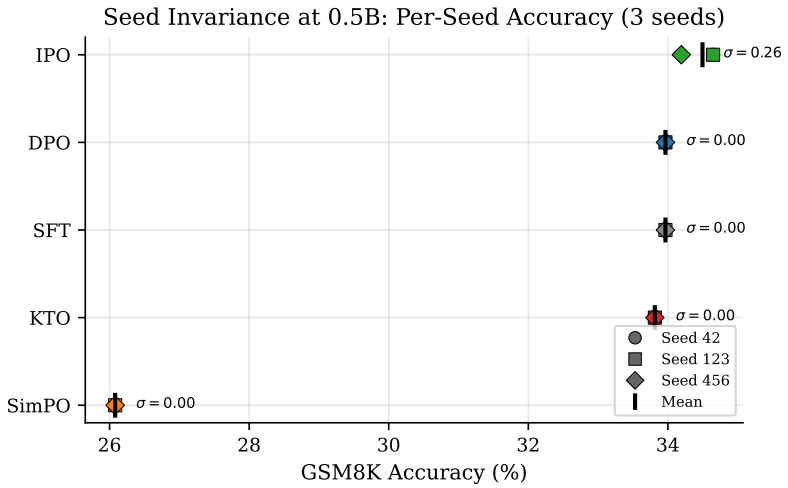

Figure 4: Training determinism at 0.5B: per-seed GSM8K accuracy for each algorithm across 3 seeds. Five of six algorithms produce identical accuracy ($\sigma = 0$).

I Compute Budget

Table 17: Compute budget breakdown on H100 80GB GPUs.

Category	**# Runs**	**Time/Run**	**GPUs**	**GPU-h**
Core 0.5B SL (5 algs $\times$ 3 seeds)	15	4–8 min	1	1.5
Core 0.5B RL (3 algs $\times$ 3 seeds)	9	$\sim$3 h	2	54
Core 1.5B SL (5 algs $\times$ 3 seeds)	15	14–27 min	1	5.0
Core 3B SL full FT (3 algs)	3	12–13 min	1	0.6
3B LoRA factorial (3 algs)	3	12–13 min	1	0.6
Core 7B SL (5 algs $\times$ 1 seed)	5	335–460 min	1	30
Core 7B RL (SGRPO, 3 epochs)	3	$\sim$480 min	2	48
7B multi-seed: SP-RFT/DPO/SimPO (N=5)	13	330–360 min	1	78
7B multi-seed: IPO/KTO (N=3)	4	335–460 min	1	27
MATH 1.5B evals (5 methods)	5	$\sim$30 min	1	2.5
DPO variants 1.5B (20 $\times$ 5 seeds)	100	14–27 min	1	30
DPO variants 7B (5 $\times$ 3 seeds)	15	330–360 min	1	90
Ablations 1.5B	$\sim$57	9–10 min	1	9
Ablations 7B (LR sweep, KL, Base init)	16	330–480 min	1–2	$\sim$100
Gemma 3 (3 algs $\times$ 3 scales $\times$ 3 seeds)	27	varies	1–2	$\sim$75
7B Full FT factorial (SP-RFT/DPO/SimPO)	3	$\sim$180 min	2	18
14B LoRA (SP-RFT/DPO/SimPO $\times$ 3 seeds)	9	$\sim$360 min	2	108
Fixed-dataset control (SP-RFT/DPO $\times$ 3 seeds)	6	14–27 min	1	2.0
1.5B LR sweep (SP-RFT/DPO $\times$ 4 LRs $\times$ 3 seeds)	18	14–27 min	1	$\sim$5.0
3B LR sweep (SP-RFT/DPO $\times$ 3 LRs $\times$ 3 seeds)	18	$\sim$25 min	1	$\sim$7.5
Gold-SFT control (1.5B $\times$ 3 + 7B $\times$ 3 seeds)	6	27–360 min	1	$\sim$20
Total training	$\sim$350	–	–	**$\sim$720**
Evaluation (GSM8K, MATH, general)	$\sim$430+	5–600 min	1	$\sim$220
Grand total				**$\sim$940 GPU-h**

The total compute ($\sim$940 GPU-hours) is modest given the breadth ($\sim$350 training runs across 2 model families, 8 scales, 4 evaluation domains) and substantially less than training a single 7B model from scratch.

J Additional Scaling Results

SGRPO at 1.5B achieves $58.00\% \pm 0.57$ (seeds: 57.47%, 57.92%, 58.61%). SL methods at 1.5B use multi-seed reruns with genuine variance (Table 18).

K Multi-Seed Verification

Table 18: Multi-seed results at 1.5B on GSM8K (3 seeds each). Rankings preserved across seeds.

Method	Original	Mean	σ	s42	s123	s456
SP-RFT	54.66	54.36	0.59	53.90	55.19	53.98
IPO	52.39	52.24	0.22	52.54	52.16	52.01
KTO	51.71	51.15	1.77	51.33	53.22	48.90
DPO	51.18	49.08	0.61	49.58	49.43	48.22
SimPO	43.44	38.67	1.78	36.85	38.06	41.09

Table 19: Multi-seed results at 7B (LoRA) on GSM8K. Core algorithms have N=5 seeds; IPO/KTO have N=3.

Method	N	Mean	σ	s42	s456	s789	s1024	s1337
DPO	5	**83.38**	0.56	83.85	82.79	83.02	84.23	83.02
SimPO	5	83.32	1.79	85.82	81.50	82.64	85.06	81.58
IPO	3	80.39	0.92	81.35	—	80.67	—	79.15
KTO	3	80.24	0.16	80.21	—	80.44	—	80.06
SP-RFT	5	77.38	1.11	76.42	79.30	76.65	76.57	77.94

Statistical tests at 7B. Welch's t-tests with Bonferroni correction ($\alpha = 0.05/10 = 0.005$):

- **DPO vs. SimPO:** $\Delta = +0.06$ pp, $p = 0.950$. Not significant; no difference detected at this sample size.
- **DPO vs. SP-RFT:** $\Delta = +6.01$ pp, $p < 0.001$. **Significant.**
- **SimPO vs. SP-RFT:** $\Delta = +5.94$ pp, $p < 0.001$. **Significant.**
- **DPO vs. KTO:** $\Delta = +3.15$ pp, $p < 0.001$. **Significant.**
- **DPO vs. IPO:** $\Delta = +2.99$ pp, $p = 0.029$. Not significant after correction.
- **SP-RFT vs. IPO:** $\Delta = -3.01$ pp, $p = 0.018$. Not significant after correction.
- **SP-RFT vs. KTO:** $\Delta = -2.86$ pp, $p = 0.006$. Not significant after correction (marginal).
- **IPO vs. KTO:** $\Delta = +0.15$ pp, $p = 0.837$. Not significant.

Three groups emerge: (1) DPO/SimPO cluster ($\sim$83.4%), (2) IPO/KTO cluster ($\sim$80.3%), (3) SP-RFT ($\sim$77.4%). The DPO/SimPO $>$ SP-RFT separation is robust ($p < 0.001$).

L Sampling-Based Evaluation

We re-evaluate 1.5B and 7B checkpoints using nucleus sampling (top-$p = 0.9$, temperature $= 0.8$) with 8 samples per problem, reporting majority-vote accuracy. Core rankings are preserved: at 7B, DPO and SimPO remain at the top, SP-RFT at the bottom. The inter-algorithm spread compresses slightly, consistent with sampling partially compensating for format compliance differences.

M Learning Rate Sensitivity at 7B

Three observations emerge. **(1)** The default 10^{-6} is optimal for both SP-RFT and DPO; no higher LR improves either algorithm, ruling out under-training as an explanation for SP-RFT's 7B performance. **(2)** DPO is *more* LR-sensitive than SP-RFT: at 10^{-5}, DPO collapses to 16.15% (one seed hits 1.2%) while SP-RFT retains 53.12%. **(3)** The ranking inversion (DPO $>$ SP-RFT) holds at the independently

Table 20: Learning rate sensitivity for SP-RFT and DPO at 7B (LoRA) on GSM8K (%). Each cell reports mean ± std over 3 seeds (SP-RFT) or 2 seeds (DPO). The default LR $= 10^{-6}$ is optimal for both algorithms; higher rates degrade performance monotonically.

LR	SP-RFT Mean	SP-RFT Seeds	DPO Mean	DPO Seeds
10^{-6} (default)	$77.38_{\pm 1.11}$	5 seeds	$83.38_{\pm 0.56}$	5 seeds
5×10^{-6}	$66.57_{\pm 5.14}$	66.3 / 60.4 / 73.0	$56.75_{\pm 8.68}$	65.4 / 48.1
10^{-5}	$53.12_{\pm 4.90}$	59.3 / 47.3 / 52.8	$16.15_{\pm 14.94}$	31.1 / 1.2
5×10^{-5}	$30.20_{\pm 7.73}$	34.1 / 37.1 / 19.4	—	—
2×10^{-4}	0.23	1 seed	—	—

optimal LR for each algorithm (10^{-6} for both), confirming it is not an artifact of hyperparameter tuning.

Note on the lower boundary. Our grid does not extend below 10^{-6} (e.g., to 5×10^{-7}). Because performance degrades *monotonically and steeply* at every LR above 10^{-6}—with DPO losing 27 pp at just $5\times$ higher—the gradient strongly favors the conservative end, and 10^{-6} already matches the recommended LoRA LR for both Qwen 2.5 and Gemma 3 in their official fine-tuning guides. Nonetheless, we cannot rule out a marginal improvement at 5×10^{-7}; if it exists, it would affect SP-RFT and DPO similarly and would not change the ranking inversion.

N Learning Rate Sensitivity at 1.5B

To directly address whether the 1.5B SP-RFT–DPO ranking is an artifact of the shared LR (10^{-6}), we sweep LR in $\{5\times10^{-7}, 10^{-6}, 5\times10^{-6}, 10^{-5}\}$ for both SP-RFT and DPO at 1.5B with 3 seeds each (24 runs total).

Table 21: Learning rate sensitivity for SP-RFT and DPO at 1.5B (full FT) on GSM8K (%). Each cell reports mean ± population std over 3 seeds. SP-RFT leads at *every* LR. DPO collapses catastrophically at $\geq 5\times10^{-6}$. At independently optimal LRs, SP-RFT@10^{-5} (56.61%) beats DPO@5×10^{-7} (52.26%) by +4.35 pp.

LR	SP-RFT Mean$_{\pm\sigma}$	SP-RFT Seeds	DPO Mean$_{\pm\sigma}$	DPO Seeds
5×10^{-7}	$53.83_{\pm 0.21}$	54.13 / 53.68 / 53.68	$\mathbf{52.26}_{\pm 0.52}$	52.46 / 52.77 / 51.55
10^{-6} (default)	$54.36_{\pm 0.59}$	55.19 / 53.90 / 53.98	$49.08_{\pm 0.61}$	49.43 / 49.58 / 48.22
5×10^{-6}	$54.74_{\pm 1.21}$	56.25 / 53.30 / 54.66	$16.00_{\pm 11.69}$	20.39 / 27.60 / 0.00†
10^{-5}	$\mathbf{56.61}_{\pm 0.48}$	55.95 / 56.79 / 57.09	$19.66_{\pm 1.81}$	20.62 / 21.23 / 17.13

†Complete collapse: model generates only degenerate output.

Four observations: **(1)** SP-RFT wins at every LR tested. **(2)** At independently optimal LRs, the gap widens to +4.35 pp. **(3)** DPO is dramatically more LR-sensitive at 1.5B than at 7B: the usable LR range for DPO shrinks from roughly one order of magnitude at 7B to half an order at 1.5B. **(4)** The 1.5B and 7B sweeps tell opposite stories about optimal LR, confirming that LR sensitivity itself is scale-dependent.

O Learning Rate Sensitivity at 3B

The 3B collapse (Section 4.1) is fully recoverable with a LR sweep. We sweep LR in $\{5\times10^{-7}, 5\times10^{-6}, 10^{-5}\}$ for both SP-RFT and DPO at 3B (full FT) with 3 seeds each (18 runs).

Three observations: **(1)** The 3B collapse is fully recoverable for SP-RFT: at 10^{-5}, it reaches 55.70%—comparable to 1.5B Instruct (54.36%). **(2)** SP-RFT wins at every LR, preserving the small-scale ranking. **(3)** DPO is dramatically more LR-sensitive than SP-RFT at 3B, mirroring the pattern at 1.5B and 7B.

Table 22: Learning rate sensitivity for SP-RFT and DPO at 3B (full FT) on GSM8K (%). Each cell reports mean ± population std over 3 seeds. SP-RFT recovers to 55.7% at 10^{-5}—comparable to 1.5B Instruct—confirming the default-LR collapse was a hyperparameter pathology. SP-RFT leads at every LR.

LR	SP-RFT Mean$_{\pm\sigma}$	SP-RFT Seeds	DPO Mean$_{\pm\sigma}$	DPO Seeds
5×10^{-7}	$17.79_{\pm0.16}$	17.82 / 17.97 / 17.59	$15.77_{\pm0.91}$	14.56 / 16.00 / 16.76
10^{-6} (default)	18.35^{b}	—	$14.44_{\pm2.01}$	16.45 / 14.44 / 12.42
5×10^{-6}	$41.80_{\pm3.84}$	40.26 / 38.06 / 47.08	$\mathbf{34.55}_{\pm4.52}$	33.81 / 29.42 / 40.41
10^{-5}	$\mathbf{55.70}_{\pm1.09}$	54.21 / 56.79 / 56.10	$16.30_{\pm7.34}$	5.91 / 21.46 / 21.53

[b] Single seed from Table 1.

Table 23: SGRPO at 7B with and without KL regularization on GSM8K (%).

Method	λ_{KL}	GSM8K (%)	Δ vs. DPO
SGRPO (default)	0	80.59	−2.79
SGRPO (regularized)	0.01	80.74	−2.64
DPO (reference)	$\beta = 0.1$	$83.38_{\pm0.56}$	—

P KL Regularization Ablation for SGRPO

Adding KL regularization yields only +0.15 pp, confirming the online–offline gap is genuine rather than a regularization artifact.

Q Base vs. Instruct Initialization at 7B

Table 24: Base vs. Instruct initialization at 7B on GSM8K (%). The spread collapses from 6.00 pp to 1.82 pp. SP-RFT leads from Base, reversing the Instruct ranking.

Algorithm	From Base (7B)	From Instruct (7B)	Δ
SP-RFT	**82.49**	$77.38_{\pm1.11}$	+5.11
DPO	80.67	$83.38_{\pm0.56}$	−2.71
SimPO	80.82	$83.32_{\pm1.79}$	−2.50
Spread	*1.82 pp*	*6.00 pp*	

From Base initialization at 7B, the ranking (SP-RFT > SimPO > DPO) differs from both the 1.5B Base ranking (SimPO > DPO > SP-RFT) and the 7B Instruct ranking (DPO > SimPO ≫ SP-RFT). A residual ranking shift between scales persists even from Base, but at ≤1.82 pp—far smaller than the 6.00 pp Instruct spread. The key finding is that initialization accounts for the majority of inter-algorithm variance (3.3× compression at 7B, 14.7× at 1.5B), even though a small scale-dependent residual remains.

R Hyperparameter Sensitivity

S 3B Flexible Extraction Analysis

Under flexible extraction, the 3B base model achieves 62.93%—comparable to the 1.5B Instruct model's strict-match performance. This reveals that the Qwen 2.5 3B Instruct model can *solve* GSM8K problems at a competitive rate but fails to produce answers in the expected format. Post-training partially addresses this: SP-RFT reduces the format gap from −52.2 pp (base) to −26.2 pp.

Table 25: β sensitivity at 1.5B on GSM8K (%). Both variants are stable across a $50\times$ range; default $\beta = 0.1$ is near-optimal.

Variant	$\beta = 0.01$	$\beta = 0.1$	$\beta = 0.5$	**Range**
DPOP	54.66	**55.19**	53.30	1.89 pp
ORPO	54.81	54.44	53.83	0.98 pp
Vanilla DPO	*49.76*$_{\pm 2.27}$ *(variant sweep)*			

T General Benchmark Results

Table 26 reports ARC-Challenge, HellaSwag, and WinoGrande accuracy across scales and architectures. At every scale and architecture, the spread is <2 pp—confirming that post-training algorithms do not meaningfully differ on these benchmarks. **This is a critical scope qualifier for our main findings:** the ranking inversion documented in this paper is specific to mathematical reasoning (GSM8K, MATH). On these general benchmarks, no inversion is observable, and algorithm choice is essentially irrelevant.

Caveat: benchmark format. ARC-C, HellaSwag, and WinoGrande are multiple-choice or perplexity-based evaluations, not open-ended generative tasks. Since our analysis identifies format compliance—a generative property—as the key differentiator at small scales (Section 4.4), the absence of algorithmic differences on non-generative benchmarks is expected and does not rule out the possibility that inversions occur on other *generative* tasks. We leave this to future work.

Table 26: General benchmark accuracy (%) after post-training. Spread remains <2 pp across all scales and architectures, confirming that algorithmic differences are task-specific.

Architecture	**Algorithm**	**ARC-C**	**HellaSwag**	**WinoGrande**	**Avg**
Gemma 1B	Base	38.7	57.7	58.9	51.8
	SP-RFT	39.1	57.9	59.1	52.0
	DPO	38.7	57.6	58.8	51.7
	SimPO	39.6	57.2	57.6	51.5
Gemma 12B	Base	61.1	81.9	74.6	72.5
	SP-RFT	60.1	81.7	74.5	72.1
	DPO	58.4	81.2	74.6	71.4
	SimPO	56.7	81.3	73.6	70.5
Qwen 7B FT	Base	55.2	80.5	71.1	68.9
	SP-RFT	56.1	80.6	71.0	69.2
	DPO	55.0	80.6	70.4	68.7
	SimPO	54.1	80.6	69.9	68.2
Qwen 14B	Base	62.4	84.4	76.0	74.3
	SP-RFT	61.9	84.5	75.2	73.9
	DPO	61.8	84.4	75.4	73.9
	SimPO	61.7	84.5	74.8	73.7